

Using VR to Elicit Empathy in Current and Future Psychiatrists for their Patients of Color

Benjamin Corriette*
Howard University

Darryl Parsons†
Howard University

Chidindu Alim‡
Howard University

Teanna Barrett**
Howard University

Tariq Cranston††
Howard University

Gloria Washington‡‡
Howard University

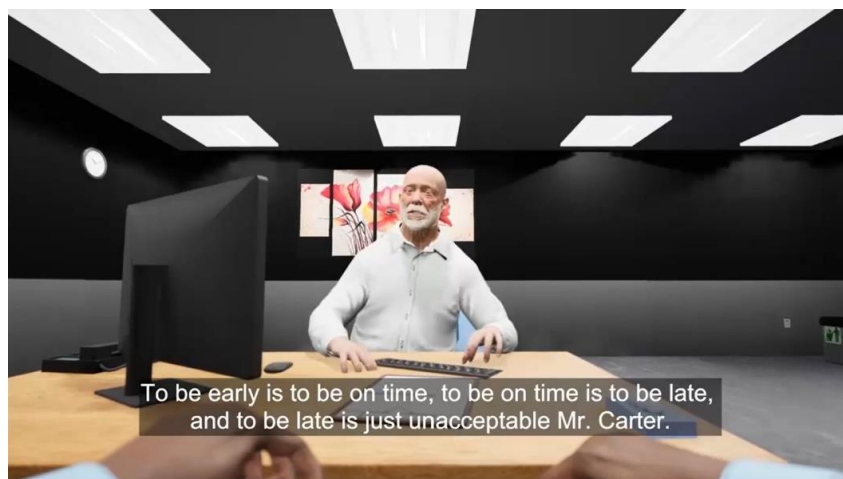


Figure 1: An example of one of Dr. Thompson's microaggressions towards Malcolm.

ABSTRACT

Individuals can now learn anti-racism and anti-discriminatory lessons in virtual reality (VR) through Cogburn's work at Columbia University. The purpose of this VR training tool is to elicit greater empathy in current and future psychiatrists toward their patients of color by observing the microaggressions that a young Black man named Malcolm Carter is subjected to by his older White male psychiatrist, Dr. Thompson. This tool is important because it will increase the quality of psychiatric care for patients of color due to the increased awareness of the inherent biases against them. The project team created this tool to explore its effects on the level of empathy that current and future psychiatrists have toward their patients of color. While developing

the tool, the team consulted with Microsoft employees who specialize in VR, a professor of psychiatry, a psychiatry resident, and a medical student, to ensure that the scenarios presented are authentic, and that the user experience is accessible to a wide audience, e.g., by including subtitles. Preliminary results reveal that the training tool is successful in eliciting empathy for Malcolm because the participants involved in testing the tool have expressed how much it is needed due to the lack of diversity within psychiatry, the general lack of trust patients of color have toward White doctors, and the negative effect that lack of trust has on the social determinants of health within communities of color. There has also been an interest in replicating this training tool for use with other races, genders, and ethnicities. Medical professionals engage in cultural competence training, but many times the training is ineffective because they often do not hear the unacceptable things they say to their patients. This VR experience has the potential to put medical professionals through the mental and emotional trauma they might inflict.

Keywords: Virtual reality, psychiatry, empathy, microaggressions, patients of color, Black men.

Index Terms: Human-centered computing—Visualization—Empirical studies in visualization; Social and professional topics—User characteristics—Race and ethnicity; Computing methodologies—Modeling and simulation—Simulation types and techniques—Simulation by animation

*email: benjamin.corriette@bison.howard.edu

†email: darryl.parsons@bison.howard.edu

‡email: chidindu.alim@bison.howard.edu

**email: teanna.barrett@bison.howard.edu

††email: tariq.cranston@bison.howard.edu

‡‡email: gloria.washington@howard.edu

1 INTRODUCTION

1.1 Background

Empathy has had many different definitions, all of which encompass similar themes. [12] points to three subprocesses of the empathetic process: emotional simulation, perspective taking, and emotion regulation.

In [13] the writers took multiple definitions coined by different authors, found their commonalities and differences, and arrived at their own definition that encompasses their best version of empathy. Their definition reads: “Empathy is an emotional response (affective), dependent upon the interaction between trait capacities and state influences. Empathic processes are automatically elicited but are also shaped by top-down control processes. The resulting emotion is similar to one’s perception (directly experienced or imagined) and understanding (cognitive empathy) of the stimulus emotion, with recognition that the source of the emotion is not one’s own”.

The parallels between [12] and [13] can be seen as the first part of the definition matches the need for an emotional response from the empathizer because of some external element. The definition in [13] does imply that the empathizer can emotionally regulate, as it says the “empathetic process is also shaped by top-down control processes” [13], and that there is a recognition that the “source of emotion is not one’s own” [13]. Finally, there is a similar aspect of perspective taking, as [13]’s definition describes that there can be a direct and imagined perception of the stimulus when empathy occurs. In VR research investigating empathy, simulations are created in which the empathizer can view a scenario as an observer of another individual, or as the individual directly from their perspective.

Cogburn’s *1000 Cut Journey* VR experience places the participant in the perspective of Michael Sterling, a young Black man, from childhood to adulthood, to witness the racism he’s dealt with throughout [14]. The motivation behind her research is to provide White liberals who have superficial surface level concepts of racism and move them towards substantive knowledge and meaningful engagement with racism. Inspired by Cogburn’s work, this VR training tool places the participant in the virtual shoes of Malcolm Carter, a young Black man who is employed as a guidance counselor at a local public school, and is visiting his psychiatrist, Dr. Thompson, an older White man. The goal of the training tool is for the participant to observe the microaggressions that Malcolm experiences during his visit (see Figure 1), so that the participant may have greater empathy toward patients of color. This tool will ultimately be used to train current and future psychiatrists.

Clouds Over Sidra is an eight-minute documentary where the viewer is guided by a young Syrian girl through her refugee camp. [5] presents this documentary to 24 participants in VR and two-dimensional formats respectively, equally split in two groups, to test how the viewing experience would affect their empathy for their tour guide Sidra and their engagement with the story. Shutte and Stilinović’s paper found that the “virtual reality experience resulted in greater engagement and a higher level of empathy for the refugee girl compared to the control condition” [5].

While *Clouds Over Sidra* focuses primarily on presence, *1000 Cut Journey* utilizes embodiment. Presence is a term used to describe “the feeling of being somewhere real when you are in VR” [6], while embodiment is a broadened definition of presence that includes being present in someone else’s body [7]. [7] also mentions that previous research indicates perspective taking is one of the best ways to foster empathy.

Ramirez in *Can Technology Help Us be More Empathetic? Racism, Empathy and Virtual Reality* claims that VR can’t make us more empathetic, and it’s important that we discuss those implications [11]. His primary reasoning for his argument is that intersectionality can’t be simulated, and to assume otherwise denies that intersectionality is a fundamental part of how we experience the world. VR is successful in providing a tangible experience for those struggling to see another person’s perspective, and by doing so is creating an assisted empathetic experience.

1.2 System Description

The VR training tool was developed in Unreal Engine, which is a digital creation platform provided by Epic Games. The project team chose Unreal Engine because of its photorealistic rendering, wealth of assets, thriving developer community and Epic Games’ commitment to democratizing game development. While developing the tool, the team consulted with Microsoft employees who specialize in VR, a professor of psychiatry, a psychiatry resident, and a medical student, to ensure that the scenarios presented are authentic, and that the user experience is accessible to a wide audience, e.g., by including subtitles.

Several three-dimensional assets were used to create the virtual environment, e.g., MetaHuman Creator, a cloud-based avatar creation tool offered by Epic Games, was used to generate the lifelike characters (see Figure 1). Character animation assets from Mixamo and other resources were used to animate the avatars, while a motion capture tool called LiveLink was used for creating animations for speaking and facial expressions. The avatars’ dialogue was then synchronized with audio recordings of voice actors during post-production.

During the latter stages of development, the idea of creating a separate “contemplation points” interactive video emerged, where the participant would pause for contemplation after each of the six scenarios and answer survey questions about how they felt regarding the microaggressions that they just experienced and how the situation should have been handled, if they were Malcolm’s provider. After each scenario, there is a 30-second narration from a psychiatrist about how psychiatry best practices should have been applied to increase the quality of Malcolm’s care. The interactive video was created in Adobe Captivate and uses a custom JavaScript action to send survey responses via Asynchronous JavaScript and XML (AJAX) requests to a custom Google Apps script that processes and stores the responses in a Google Sheet. Each participant’s survey responses are associated with a unique ID for data analysis.

1.3 System Evaluation

The VR training tool will be piloted by a cohort of medical students studying psychiatry during the Spring 2023 semester. Ultimately, the project team plans to demonstrate this tool at psychiatry conventions, symposiums, etc., so that more current and future psychiatrists can experience it. Using this tool will be an impactful way of supplementing cultural competency training for psychiatrists.

During the pilot, three participants at a time will be scheduled for one-hour time slots. They will be seated at separate Windows 10 computer stations, each equipped with a VR head-mounted display (HMD). The project team will have two Oculus Rift HMDs and an HTC Vive HMD (see Figure 2) for participants to use during the pilot. The Oculus Rift is a newer model than the HTC Vive, but the project team does not anticipate any significant differences in user experience. For those who want to experience

the training tool, but do not have access to a VR HMD, there will be a 360-degree video uploaded to YouTube that they can access.

To date, the project team has shared the training tool with a small number of interested third parties, and their feedback has been overwhelmingly positive. Malcolm's experience is something that many of them can relate to based on their past experiences as people of color receiving healthcare.

In the next section, the theory behind this VR training tool will be discussed, followed by a discussion of its limitations, its similarities and differences with the related literature, planned future work, and finally, the conclusion.



Figure 2: HTC Vive VR Head-mounted display.

2 THEORY

1000 Cut Journey, *Driving While Black*, and *I am a Man* are all VR experiences with the common goal of fostering empathy within its audience to understand how racism can affect the Black community. Similarly, this VR training tool aims to show another facet of the Black experience and how racial bias can come from those who have sworn to do no harm to their patients.

In this section, we will discuss how VR has been used to foster empathy in the following areas: healthcare, education, and criminal justice/law enforcement. Through this section, we will examine how each paper went about evaluating empathy within their application and how effective it was at making participants more empathetic to a certain demographic.

2.1 VR in Healthcare

A leading issue revealed by studies showed that “general healthcare professionals had significant shortcomings in knowledge and attitude pertaining to psychiatric disorders, which hinder the provision of adequate and appropriate services” [1]. It is unfortunate that those with mental illness often can't trust the health professionals to understand their needs and is a problem that needs to be fixed expeditiously. Kavanagh et al. found after evaluating six articles that VR simulation can improve participants' empathy, attitude, and knowledge relating to mental health. The advancement of said VR technology has led to more realistic simulations and is “suggested to have high flexibility in providing systematic experiential training that optimizes the degree of transfer of training to individual's reality perceptions” [1]. Two studies of the six examined supported that VR simulation increases the user's empathy. Additionally, two studies

showed improvement in users' attitudes towards mental illness and five studies unanimously concluded that VR simulation provides “improvement in awareness of disease, symptom identification and skill competence [for users]” [1]. These findings are hopeful, as we want to train psychiatrists to understand how medical racism can affect their patients, and this VR training tool can be effective in that process.

2.2 VR in Education

The highlight of [6] is that VR can be used for educational purposes that benefit the users while doing minimal harm compared to its real-world counterparts. An example they gave is simulating surgery for new and upcoming students instead of having them do it on a live human subject. This can be compared to our desire to teach how medical racism can harm Black people who experience it. By having an interactive simulation, we minimize the trauma put on Black individuals having to potentially relive past trauma. Freina et al. also speak on how from their survey they found that published papers on VR technology within education fell into two groups: “those aimed at adult vocational training and those that refer to high school and university education” [6].

2.3 VR in Criminal Justice/Law Enforcement

Though social distance has taken on a new meaning due to the COVID-19 pandemic, social distance in [10] refers to the emotional closeness to other individuals. Kim argues that VR technology can close the gap in social distance, thereby making people more empathetic. Kim states that “VR training can be reciprocal as well. While it offers community members an opportunity to develop empathy towards police officers through shared experiences, it can simultaneously offer police officers the opportunity to develop empathy toward community members as well.” [10]. Though it discusses how community members can safely experience what police officers may endure on the job, the focus will be on potential uses in the other direction, e.g., VR could be used to foster empathy by showing police officers how police brutality affects the communities they serve.

3 DISCUSSION

3.1 Limitations

Participants who do not have a VR HMD must view the training video on YouTube. The video is in 360-degree format, so the participant must use a mouse and keyboard to interact with it. This limitation will be an issue for participants with fine motor skill challenges, e.g., dyspraxia.

3.2 Similarities and Differences

3.2.1 Similarities

This training tool is similar to VR used in education, e.g., Cogburn's *1000 Cut Journey*, because it aims to teach individuals about the biases faced by marginalized groups while minimizing the negative effects of Black people having to act out these marginalizing situations [2], [6], [14].

It is also similar to VR used in healthcare, because it fosters empathy for a marginalized group, whereas previous VR experiences in healthcare focused on healthcare practitioners and their patients, regardless of their socioeconomic status.

3.2.2 Differences

This training tool is different from VR used in criminal justice because the research doesn't address the physical safety of people

but addresses their mental health and the biases that may they encounter while receiving healthcare.

3.3 Future Work

The collection of the participants' biometric data is being considered because it will help the researchers to understand the physiological responses to the training, which can then be correlated to the textual data collected. There is also the possibility of replicating this tool to train psychiatrists about other underrepresented races, genders, and ethnicities.

CONCLUSION

In VR research investigating empathy, simulations are created in which the empathizer can view a scenario as the individual directly from their perspective, e.g., in Cogburn's *1000 Cut Journey* that places the participant in the perspective of Michael Sterling, a young Black man, with the goal of fostering empathy for him via the racism he experiences. Similarly, this training tool aims to show how racial bias negatively impacts the care that Malcolm Carter, also a young Black man, receives from his older White male psychiatrist, with the intention of fostering empathy in current and future psychiatrists for their patients of color.

Due to the lack of diversity within psychiatry, and the historical lack of trust that people of color have toward White doctors, this VR training tool can play a vital role in helping to improve the quality of their psychiatric care.

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